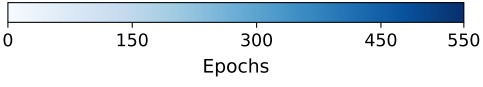
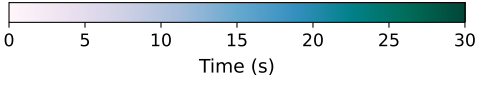
A Completed training epochs

	raw			TC			DESeq		
	1:5	1:10	1:50	1:5	1:10	1:50	1:5	1:10	1:50
BLCA	292	280	162	259	254	167	366	332	152
BRCA	474	433	270	376	508	173	454	350	147
COAD	377	260	177	305	217	185	276	274	188
ESCA	184	132	83	165	82	138	223	208	154
HNSC	295	434	115	177	265	220	344	301	233
KICH	78	102	97	89	98	100	100	70	120
KIRC	213	273	105	281	122	126	377	376	97
KIRP	149	185	137	135	123	102	312	331	128
LAML	169	157	127	205	204	104	157	283	124
LIHC	255	245	92	286	272	127	287	248	176
LUAD	514	498	212	362	192	92	247	310	121
LUSC	424	221	163	280	232	173	238	210	230
OV	433	403	198	306	227	87	250	334	243
PAAD	145	189	133	109	159	145	89	96	151
PCPG	178	150	141	200	149	86	185	184	85
PRAD	281	232	72	243	283	71	286	148	143
READ	123	115	71	86	84	101	76	99	71
SARC	160	227	108	269	170	107	211	137	96
SKCM	292	287	251	250	321	230	283	240	237
STAD	286	305	136	274	203	181	306	260	99
TGCT	115	111	101	135	96	138	99	139	86
THCA	301	193	116	368	123	271	534	294	222
THYM	147	139	161	161	121	77	182	148	139
UCS	96	131	118	79	148	109	78	95	81



B Training and generation time

	raw			TC			DESeq		
	1:5	1:10	1:50	1:5	1:10	1:50	1:5	1:10	1:50
BLCA	13.1	10.3	6.2	10.2	10.9	6.7	13.7	12.2	6.0
BRCA	22.1	20.4	15.4	21.5	27.1	9.8	23.1	17.7	7.9
COAD	14.7	9.4	6.7	11.8	8.7	7.1	10.1	10.0	6.9
ESCA	6.2	4.4	3.0	5.8	3.4	5.0	8.4	7.1	5.5
HNSC	11.5	17.1	4.7	6.9	10.2	9.1	14.0	12.2	9.3
KICH	2.7	3.5	3.2	3.0	3.8	3.6	3.7	2.5	4.1
KIRC	7.8	10.3	4.0	10.4	4.6	5.4	15.8	14.2	3.9
KIRP	4.7	5.9	4.4	4.3	4.0	3.3	11.5	10.8	4.3
LAML	5.4	4.8	3.9	6.3	6.2	3.2	4.8	8.8	4.1
LIHC	10.6	9.9	4.0	11.5	10.8	5.2	11.1	10.7	7.6
LUAD	17.1	16.5	7.3	11.9	6.3	3.2	8.1	10.1	4.0
LUSC	15.9	8.1	6.3	10.5	8.8	6.6	8.8	7.9	9.6
OV	18.8	15.1	7.8	11.7	9.2	3.7	9.8	13.2	9.5
PAAD	5.4	6.8	5.1	3.8	5.5	5.0	3.2	3.4	5.1
PCPG	5.3	4.5	4.7	6.6	4.6	2.8	5.6	5.6	2.7
PRAD	8.9	7.3	2.5	9.7	12.0	2.6	10.3	5.3	4.9
READ	4.6	4.0	2.5	3.0	3.6	4.5	3.6	4.4	3.0
SARC	5.8	8.2	3.6	8.8	5.7	3.6	7.3	4.9	3.5
SKCM	12.7	13.2	12.0	10.3	13.2	9.5	11.6	12.7	11.9
STAD	13.9	13.5	6.3	10.8	8.3	7.1	12.0	10.1	4.2
TGCT	4.4	4.2	3.8	4.9	3.5	4.9	3.5	6.9	3.7
THCA	13.1	8.3	5.3	15.8	5.5	10.9	21.4	11.5	10.5
THYM	5.8	5.8	6.2	6.1	4.9	3.1	6.4	5.4	4.8
UCS	3.1	4.5	3.5	2.4	4.4	3.3	2.3	2.8	2.5



Columns: normalization and cVAE reconstruction-to-KL loss ratio